

Two-sided Pimsner–Popa bases: regular and irreducible depth-two subfactors

Literature-implied partial answer for arXiv:1710.00285

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Status. Affirmative in two large later-literature subcases. Every finite-index regular inclusion of II_1 factors has a two-sided Pimsner–Popa basis [2]. Every finite-index irreducible depth-two inclusion has, more strongly, a two-sided orthonormal basis [3]. These results answer the source question for regular subfactors and for all irreducible depth-two subfactors, including nonregular Kac-algebra examples. They do not answer the question for an arbitrary finite-index irreducible (equivalently here, extremal) subfactor.

Source question

For $N \subset M$, the source calls a finite set $\{\lambda_i\} \subset M$ a right Pimsner–Popa basis when

$$x = \sum_i \lambda_i E_N(\lambda_i^* x) = \sum_i E_N(x \lambda_i) \lambda_i^* \quad (x \in M).$$

It is a left basis when $\{\lambda_i^*\}$ is a right basis, and it is two-sided when both conditions hold. Remark 3.14 asks whether every finite-index (irreducible) subfactor admits such a two-sided basis [1, p. 8].

Motivated by [SW94] we try to investigate the angle $\alpha_M^N(P, Q)$ in terms of Pimsner–Popa basis [PP86]. In this paper, by Pimsner–Popa basis we mean a ‘right basis’. Thus the condition for the set $\{\lambda_i : i \in I\} \subset M$ (for some finite index set I) to be a right basis for M/N would be $\sum_{i=1}^n \lambda_i e_1 \lambda_i^* = 1$ or equivalently, $x = \sum_{i=1}^n E_N(x \lambda_i) \lambda_i^* = \sum_{i=1}^n \lambda_i E_N(\lambda_i^* x)$ for all $x \in M$. The set $\{\lambda_i : i \in I\}$ will be called a ‘left basis’ for M/N if $\{\lambda_i^* : i \in I\}$ is a right basis.

Remark 3.14. A set $\{\lambda_i : i \in I\} \subset M$ is called a two-sided basis for M/N , if it is both a left basis and a right basis. It is an open question whether any finite index, (irreducible) subfactor has a two-sided basis.

Source evidence: arXiv:1710.00285, PDF page 8.

Later affirmative subcases

Theorem 1 (Bakshi–Gupta, regular case [2]). *If $N \subset M$ is a regular subfactor of type II_1 with finite Jones index, then M/N admits a two-sided Pimsner–Popa basis.*

Theorem 2 (Bakshi–Gupta, depth-two case [3]). *Let $N \subset M$ be a finite-index depth-two inclusion of II_1 factors with simple relative commutant $N' \cap M$. Then M/N admits a two-sided orthonormal basis. In particular, this holds for every finite-index irreducible depth-two inclusion.*

The second theorem is not subsumed by regularity. The answering paper points out that an outer action of a finite-dimensional Kac algebra that is not a group algebra gives a nonregular irreducible depth-two example covered by the theorem.

Proof intuition

The regular proof factors the inclusion through

$$R = N \vee (N' \cap M).$$

The finite-dimensional algebra $N' \cap M$ supplies a two-sided basis for R/N . The generalized Weyl-group normalizers supply a two-sided basis for M/R . Because those normalizers preserve both N and R , the two bases can be patched: if $\{\lambda_i\}$ is the first and $\{\mu_j\}$ the second, then $\{\mu_j \lambda_i\}_{i,j}$ is two-sided for M/N .

For depth two, the Jones tower provides a nondegenerate commuting square

$$\begin{array}{ccc} M_1 & \subset & M_2 \\ \cup & & \cup \\ M' \cap M_1 & \subset & M' \cap M_2. \end{array}$$

Consequently, a left or right basis for the bottom finite-dimensional inclusion lifts to the top inclusion. Simplicity of $N' \cap M$ and the depth-two basic-construction identities identify the bottom inclusion with

$$M_n(\mathbb{C}) \subset M_n(\mathbb{C}) \otimes Q$$

for a finite-dimensional C^* -algebra Q . A two-sided orthonormal basis of $Q/\mathbb{C}$ then gives one for the bottom row and hence for M_2/M_1 . A two-step downward basic construction transfers the conclusion back to M/N .

Exact scope and obstruction to the obvious upgrade

The depth-two proof uses two facts that are absent at higher depth: equality of the inclusion-matrix norm with the Jones index makes the commuting square nondegenerate, and the relevant relative-commutant inclusion splits as a tensor factor $M_n(\mathbb{C}) \subset M_n(\mathbb{C}) \otimes Q$. At higher depth neither statement follows from irreducibility or extremality alone. Thus iterating the Jones tower does not upgrade [3] to arbitrary finite depth.

The source's separate conjecture that the number of minimal intermediate subfactors grows polynomially in $[M : N]$ is not addressed by either basis paper. This packet makes no claim about that conjecture.

Provenance. This is a literature-implied answer to explicit proper subcases, not a new proof from the run. The 2019 regular theorem includes one source author; the 2021 paper explicitly presents its depth-two theorem as partial progress on Jones' general two-sided-basis question. Exact arXiv-id, title, author, and core-keyword searches through 13 August 2026 found no later arXiv theorem settling the unrestricted question.

References

- [1] K. C. Bakshi, S. Das, Z. Liu, and Y. Ren, *An angle between intermediate subfactors and its rigidity*, arXiv:1710.00285 (2017). See Remark 3.14 on PDF page 8.
- [2] K. C. Bakshi and V. P. Gupta, *On orthogonal systems, two-sided bases and regular subfactors*, arXiv:1904.05612 (2019). See Theorem 3.10 on PDF page 13.
- [3] K. C. Bakshi and V. P. Gupta, *A few remarks on Pimsner–Popa bases and regular subfactors of depth 2*, arXiv:2102.01462 (2021). See Theorem 3.14 and Corollary 3.15 on PDF pages 7–9.